# C. Colorful Table

time limit per test 1 second
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1300 | Tags: binary search, data structures, dp, implementation, math, two pointers |
https://codeforces.com/problemset/problem/1870/C

You are given two integers $n$ and $k$. You are also given an array of integers $a_1, a_2, \ldots, a_n$ of size $n$. It is known that for all $1 \le i \le n$, $1 \le a_i \le k$.

Define a two-dimensional array $b$ of size $n \times n$ as follows: $b_{i,j} = \min(a_i, a_j)$. Represent array $b$ as a square, where the upper left cell is $b_{1,1}$, rows are numbered from top to bottom from 1 to $n$, and columns are numbered from left to right from 1 to $n$. Let the color of a cell be the number written in it (for a cell with coordinates $(i, j)$, this is $b_{i,j}$).

For each color from 1 to $k$, find the smallest rectangle in the array $b$ containing all cells of this color. Output the sum of width and height of this rectangle.

## Input

The first line contains a single integer $t$ ($1 \le t \le 10^4$) — the number of test cases. Then follows the description of the test cases.

The first line of each test case contains two integers $n$ and $k$ ($1 \le n, k \le 10^5$) — the size of array $a$ and the number of colors.

The second line of each test case contains $n$ integers $a_1, a_2, \ldots, a_n$ ($1 \le a_i \le k$) — the array $a$.

It is guaranteed that the sum of the values of $n$ and $k$ over all test cases does not exceed $10^5$.

## Output

For each test case, output $k$ numbers: the sums of width and height of the smallest rectangle containing all cells of a color, for each color from 1 to $k$.

## Example

| Input |
| --- |
| 5<br>2 1<br>1 1<br>2 2<br>1 2<br>3 5<br>3 2 4<br>4 2<br>1 2 1 2<br>5 3<br>1 2 3 2 1 |

**Output**

```
4
4 2
0 6 6 2 0
8 6
10 6 2
```

## Note

In the first test case, the entire array *b* consists of color 1, so the smallest rectangle for color 1 has a size of 2 × 2, and the sum of its sides is 4.

In the second test case, the array *b* looks like this:

1112

One of the corner cells has color 2, and the other three cells have color 1. Therefore, the smallest rectangle for color 1 has a size of 2 × 2, and for color 2 it is 1 × 1.

In the last test case, the array *b* looks like this:

111111222112321122211111 1